

Eindhoven University of Technology

AI-Supported Sustainability Self-Assessment for TU/e Research

ENSU – Written Assignment Stage 1

Course: 5ARF0 Ethics of AI: Theory and Practice
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Main-text count excludes headings, tables, figures, references, appendices and draft placeholders.

1 Introduction and Motivation

1.1 Topic, Motivation, Values and Needs

[TODO: Define the sustainability problem in TU/e research. Explain why an AI-supported tool may help. Cite evidence.]

Values and needs. [TODO: Introduce stakeholder needs and the main values: sustainability, transparency, accountability, fairness, and human oversight.]

1.2 Objectives and Key Terms

[TODO: State one project aim and one research question. Define the essential terms.]

1.3 Context and Scope

[TODO: Specify users, research activity, decision point, setting, and non-goals.]

1.4 Tools, Methods and Frameworks

[TODO: Explain why Design for Values or Value Sensitive Design fits the aim. Introduce conceptual, empirical, and technical investigations with references.]

Logic: research context → stakeholder need → value → requirement → tool feature → success indicator.

2 Initial Results

2.1 Preliminary Literature Search

[TODO: Report databases, search terms, dates, selection criteria, number of sources, and main evidence gaps.]

Table 1: Literature-search log.

Source	Type	Main finding	Need or value	Design implication
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]

Preliminary findings. [TODO: Synthesize researcher needs, current assessment methods, AI opportunities, known risks, and evidence gaps.]

2.2 Design Strategy

[TODO: Describe the steps from evidence and stakeholder input to requirements, mock-up, expert review, evaluation, and revision.]

2.3 Initial Requirements and Constraints

[TODO: Separate functional, non-functional, ethical requirements, constraints, and non-goals.]

Table 2: Requirement traceability matrix.

ID	Stakeholder need	Value	Requirement	Tool feature	Success indicator
R1	[TODO: Need]	Sustainability	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R2	[TODO: Need]	Transparency	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R3	[TODO: Need]	Human over-sight	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R4	[TODO: Need]	[TODO: Value]	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]

2.4 Anticipated Function and Purpose of the Tool

[TODO: Describe the user task, inputs, AI function, outputs, next human decision, and connection to the current TU/e self-assessment tool. State how experts can modify it.]

Figures 1 and 2 present a design proposal for group confirmation, rather than implemented functionality. AI would structure descriptions and draft source-supported suggestions; a calculation module would apply explicit formulas. Missing inputs would be returned for clarification, with unsupported estimates withheld if the information remains unavailable.

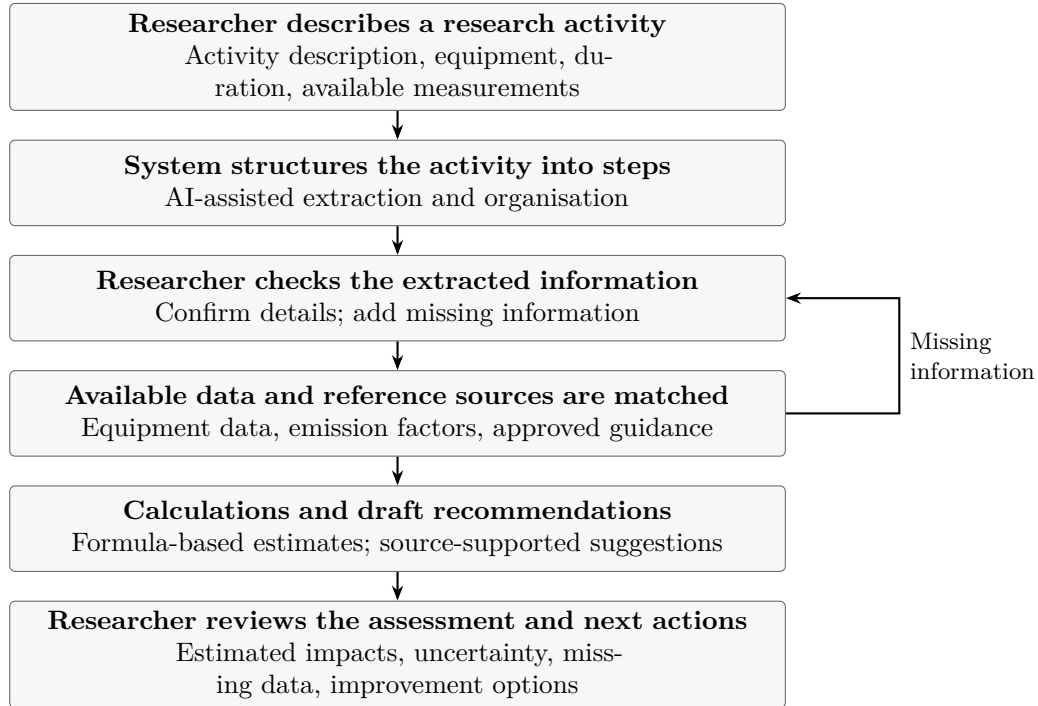


Figure 1: Proposed workflow for AI-supported sustainability self-assessment. Researchers verify activity information before reviewing estimates and source-supported recommendations.

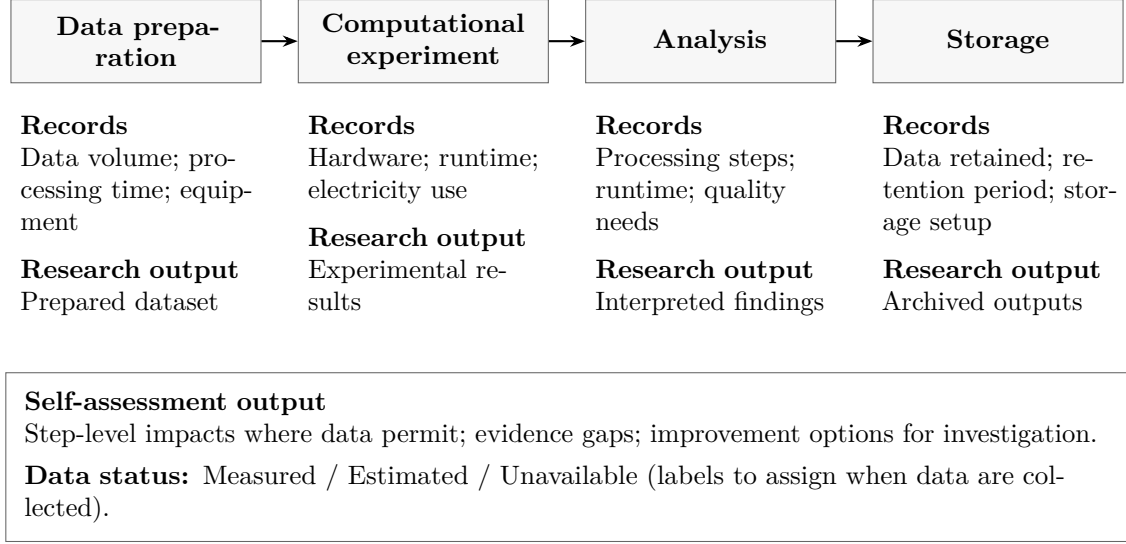


Figure 2: Illustrative computing activity and its assessment records. Research outputs are shown separately from self-assessment outputs; no measurements are assumed.

Illustrative assessment boundary: selected computing activities. Hardware manufacturing and shared infrastructure require additional data.

2.5 Projected Users, Stakeholders and Values

[TODO: Identify users, affected groups, governors, experts, and their roles in the design.]

Table 3: Stakeholder and value map.

Stakeholder	Need	Value	Benefit or harm	Design role
TU/e researchers	[TODO: Need]	[TODO: Value]	[TODO: fit/harm]	[TODO: Role]
[TODO: Stakeholder]	[TODO: Need]	[TODO: Value]	[TODO: fit/harm]	[TODO: Role]
[TODO: Stakeholder]	[TODO: Need]	[TODO: Value]	[TODO: fit/harm]	[TODO: Role]

3 Discussion

3.1 Ethical Issues and Value Conflicts

What should our tool count as sustainable research? Hasselbalch and van Wynsberghe [2025] connect sustainability to power within sociotechnical systems. For us, this raises the question of who sets the assessment criteria. Prioritising electricity consumption could penalise an intensive experiment despite its scientific value. We would therefore present environmental indicators alongside researchers’ explanations of purpose, comparing activities only where objectives and research quality are comparable. The criteria still need agreement with the client. Researchers should be able to explain why a demanding step is necessary, rather than receive a lower score without context. This would make disagreements visible instead of hiding them inside an aggregate rating. The client would still need to decide how competing considerations are weighed and whether comparisons across disciplines are appropriate for the intended use of the assessment

tool.

Numerical estimates could also acquire more authority than their evidence warrants. Lacoste et al. [2019] show how hardware, runtime and electricity supply affect emissions from machine learning training; their method does not validate assessments of every research activity. In our tool, apparent differences may be smaller than calculation uncertainty. Results should distinguish measured from estimated values, flag missing information and identify sources and boundaries. Researchers must be able to correct inputs, although this cannot ensure that they question the output.

The tool’s own resource use also needs justification. Lin [2023] argues for transparency about AI’s energy consumption. We would use fixed formulas where available and reserve AI for interpreting descriptions and drafting supported suggestions. Its contribution should be tested against a structured non-AI assessment, accounting for computation and human review. Environmental benefit remains an untested hypothesis.

3.2 Design Barriers and Challenges

A manageable assessment boundary is the first practical challenge. Shared equipment and computing services make consumption difficult to attribute. Mapping one representative workflow with the client would establish which records researchers can supply. An estimate of equipment operation must state its exclusions; it cannot stand for a complete lifecycle assessment. Replacement advice would also need evidence about manufacturing impacts, not merely operating savings.

Recommendations depend on evidence that requires maintenance. Reference data and guidance need sources and review dates, with version histories so earlier assessments remain traceable. Applying the institutional perspective of Hasselbalch and van Wynsberghe [2025], TU/e would need someone authorised to correct or withdraw outdated advice. The responsible role and available resources remain unresolved. This responsibility matters because researchers cannot reasonably verify every equipment parameter themselves. A source link alone does not establish that guidance remains appropriate for their activity.

Finally, intended users should compare AI-assisted feedback with a structured checklist on the same cases, checking whether advice follows from the evidence and recording omissions and correction effort. This would test whether the assistance is useful enough to justify further development.

Table 4: Anticipated ethical risks and proposed responses.

Risk	Stakeholders	Values	Response	Limitation
Privacy	Researchers	Privacy; autonomy	Request only necessary activity data	Descriptions may reveal sensitive work
Unclear AI advice	Researchers	Transparency; accountability	Show sources and uncertainty	Sources may be misapplied
Greenwashing	TU/e; researchers	Sustainability; trust	State boundaries and exclusions	Partial estimates omit wider impacts
Rebound effects	TU/e; affected communities	Environmental impact	Track total use alongside efficiency	Future demand is uncertain
Unfair recommendations	Research groups	Fairness	Compare similar research objectives	Criteria remain contestable
Reduced human control	Researchers	Human oversight	Require input checks and final review	Users may accept advice uncritically

References

- Antoine Berthelot, Eddy Caron, Maxime Jay, and Laurent Lefèvre. Understanding the environmental impact of generative ai services. *Communications of the ACM*, 68(7):46–53, 2025. doi: 10.1145/3725984.
- Gry Hasselbalch and Aimee van Wynsberghe. Artificial intelligence, power, and sustainability. In Nathalie A. Smuha, editor, *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence*, pages 117–130. Cambridge University Press, 2025.
- Alexandre Lacoste, Alexandra Luccioni, Victor Schmidt, and Thomas Dandres. Quantifying the carbon emissions of machine learning. arXiv preprint arXiv:1910.09700, 2019.
- Patrick K. Lin. The cost of training a machine: Lighting the way for a climate-aware policy framework that addresses artificial intelligence’s carbon footprint problem. *Fordham Environmental Law Review*, 34(2), 2023. URL <https://ir.lawnet.fordham.edu/elr/vol34/iss2/1/>.

A Documentation of Process

[TODO: Record meetings, research, design work, writing work, decisions, and evidence.]

Table 5: Project process log.

Date	Participants	Activity	Output or decision	Next action
18 Sep 2026	Yehao Chen; AI writing assistance	Revise discussion; diagrams	Dis-draft Proposed workflow; revised text	Group review of scope and wording
[TODO: Date]	[TODO: Names]	[TODO: Activity]	Activ-[TODO: Output]	[TODO: Action]

B Project Planning

Table 6: Proposed remaining project schedule (2026).

Date	Activity	Owner	Deliverable
18 Sep, 23:59	Integrate and submit Phase 1	[TODO: Owner]	Phase 1 report
22–29 Sep	Confirm scope and evidence needs during coaching	[TODO: Owner]	Revised requirements
30 Sep–10 Oct	Develop mock-up; prepare evaluation and practice pitch	[TODO: Owner]	Mock-up and evaluation plan
11 Oct, 23:59	Submit Phase 2 draft for peer review	[TODO: Owner]	Draft report
16 Oct, 23:59	Complete individual peer review	Each member	Peer feedback
17–20 Oct	Revise and record pitch	[TODO: Owner]	Video pitch: 20 Oct, 23:59
23 Oct, 23:59	Submit Phase 2 final version	[TODO: Owner]	Final report
24 Oct, 23:59	Complete groupmate evaluation	Each member	Individual evaluation